Python SQLite3

Database Connection

Code:

```
import sqlite3


try:
    sqliteConnection = sqlite3.connect('sql.db')
    cursor = sqliteConnection.cursor()
    print('DB Init')


    query = 'SELECT sqlite_version();'
    cursor.execute(query)


    result = cursor.fetchall()
    print('SQLite Version is {}'.format(result[0][0]))


    cursor.close()


except sqlite3.Error as error:
    print('Error occurred -', error)


finally:
    if sqliteConnection:
        sqliteConnection.close()
        print('SQLite Connection closed')
```

Output:

```
DB Init
SQLite Version is 3.50.4
SQLite Connection closed
```

DML Commands:

Code:

```
import sqlite3


conn = sqlite3.connect('geeks.db')
cursor = conn.cursor()


cursor.executescript("""
DROP TABLE IF EXISTS STUDENT;


CREATE TABLE STUDENT(
    NAME VARCHAR(255),
    CLASS VARCHAR(255),
    SECTION VARCHAR(255)
);


INSERT INTO STUDENT VALUES ('Raju', '7th', 'A');
INSERT INTO STUDENT VALUES ('Shyam', '8th', 'B');
INSERT INTO STUDENT VALUES ('Baburao', '9th', 'C');
""")


conn.commit()


print("Data Inserted in the table:")
cursor.execute("SELECT * FROM STUDENT")
for row in cursor.fetchall():
    print(row)
cursor.execute('''UPDATE STUDENT SET CLASS = '10th' WHERE NAME = 'Raju';''')
print('\nAfter Updating...\n')


print("STUDENT Table: ")
data = cursor.execute('''SELECT * FROM STUDENT''')
```

```
for row in data:
    print(row)
query = "SELECT * FROM STUDENT where NAME = 'Raju'"
print("Raju Details: ")
cursor.execute(query)
for row in cursor.fetchall():
    print(row)
conn.close()

conn.close()
```

Output:

```
Data Inserted in the table:
('Raju', '7th', 'A')
('Shyam', '8th', 'B')
('Baburao', '9th', 'C')

After Updating...

STUDENT Table:
('Raju', '10th', 'A')
('Shyam', '8th', 'B')
('Baburao', '9th', 'C')
Raju Details:
('Raju', '10th', 'A')
```

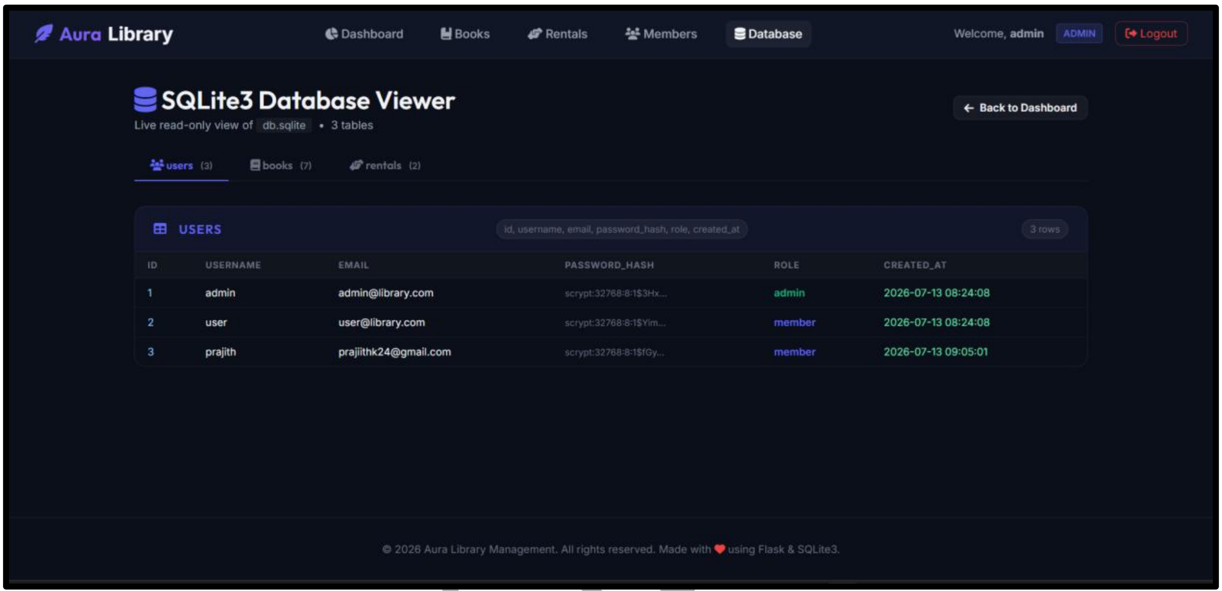
Application Using sqlite3 and flask:

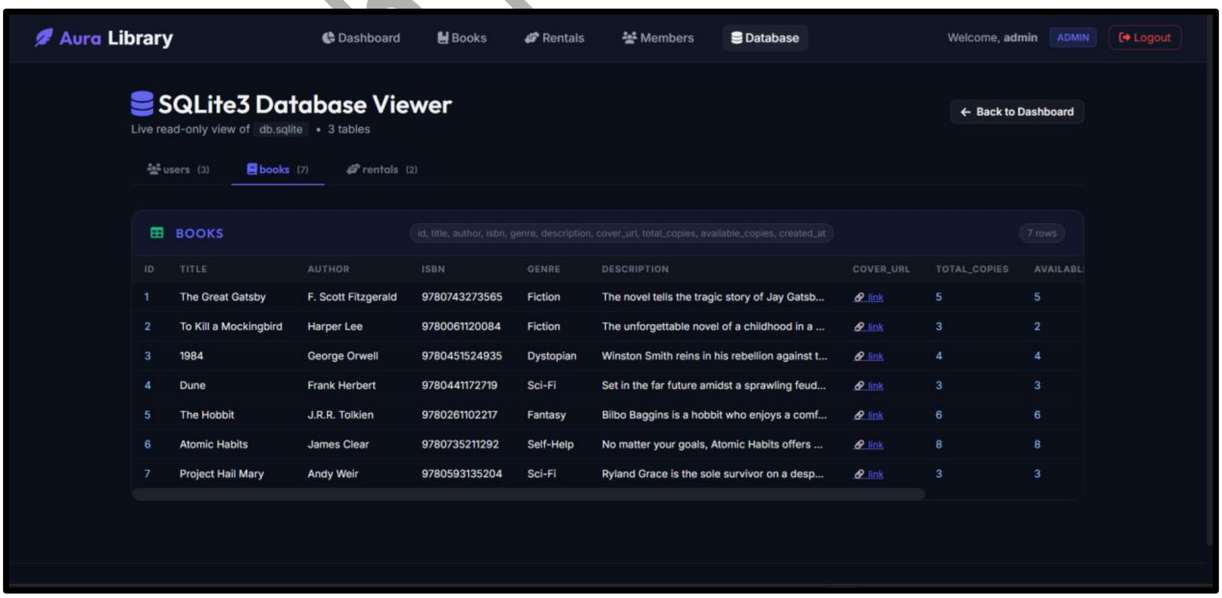
Code:

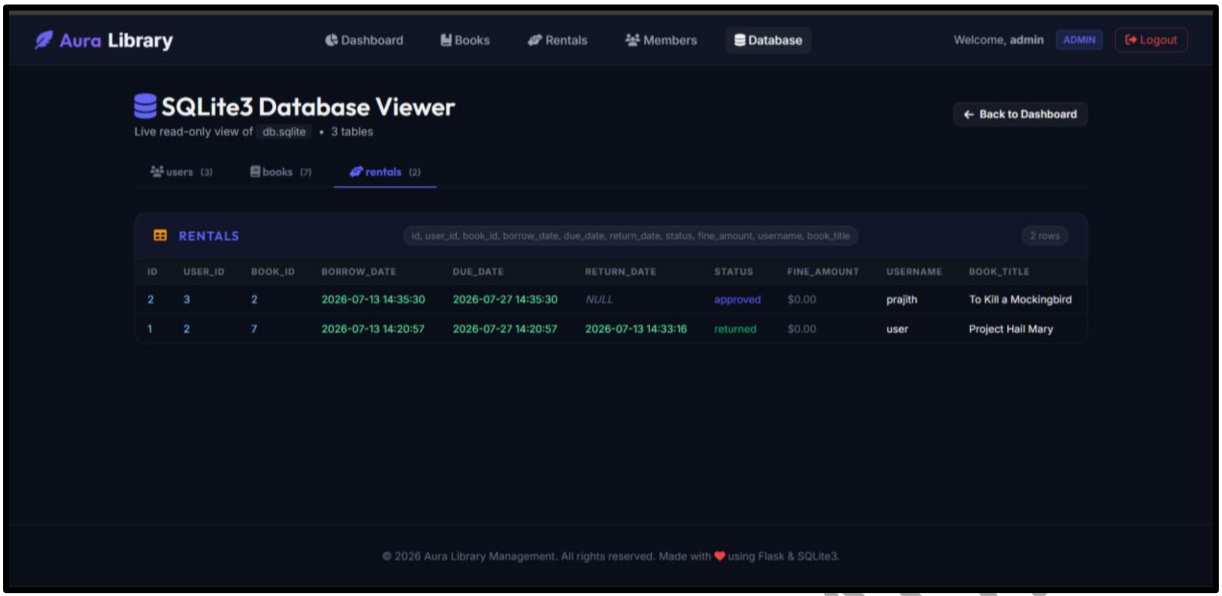
```
CREATE TABLE IF NOT EXISTS users (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    username TEXT UNIQUE NOT NULL,
    email TEXT UNIQUE NOT NULL,
    password_hash TEXT NOT NULL,
    role TEXT NOT NULL CHECK(role IN ('admin', 'member')),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE IF NOT EXISTS books (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    title TEXT NOT NULL,
    author TEXT NOT NULL,
    isbn TEXT UNIQUE NOT NULL,
    genre TEXT,
    description TEXT,
    cover_url TEXT,
    total_copies INTEGER NOT NULL DEFAULT 1,
    available_copies INTEGER NOT NULL DEFAULT 1,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE IF NOT EXISTS rentals (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    user_id INTEGER NOT NULL,
    book_id INTEGER NOT NULL,
    borrow_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    due_date TIMESTAMP NOT NULL,
    return_date TIMESTAMP,
    status TEXT NOT NULL CHECK(status IN ('pending', 'approved', 'returned', 'rejected', 'overdue')),
```

```
    fine_amount REAL DEFAULT 0.0,
    FOREIGN KEY(user_id) REFERENCES users(id) ON DELETE CASCADE,
    FOREIGN KEY(book_id) REFERENCES books(id) ON DELETE CASCADE
);
```

Output:






Aura Library
Dashboard
Books
Rentals
Members
Database
Welcome, admin
ADMIN
Logout
SQLite3 Database Viewer
Live read-only view of db.sqlite • 3 tables
Back to Dashboard
users (3)
books (7)
rentals (2)
RENTALS
id, user_id, book_id, borrow_date, due_date, return_date, status, fine_amount, username, book_title
2 rows
ID USER_ID BOOK_ID BORROW_DATE DUE_DATE RETURN_DATE STATUS FINE_AMOUNT USERNAME BOOK_TITLE
2 3 2 2026-07-13 14:35:30 2026-07-27 14:35:30 NULL approved $0.00 prajith To Kill a Mockingbird
1 2 7 2026-07-13 14:20:57 2026-07-27 14:20:57 2026-07-13 14:33:16 returned $0.00 user Project Hail Mary
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